



<http://algs4.cs.princeton.edu>

ALGORITHMS, PARTS I AND II

- ▶ *overview*
- ▶ *why study algorithms?*
- ▶ *resources*

Course overview

What is this course?

- Intermediate-level survey course.
- Programming and problem solving, with applications.
- **Algorithm:** method for solving a problem.
- **Data structure:** method to store information.

topic	data structures and algorithms	
data types	stack, queue, bag, union-find, priority queue	part 1
sorting	quicksort, mergesort, heapsort	
searching	BST, red-black BST, hash table	
graphs	BFS, DFS, Prim, Kruskal, Dijkstra	part 2
strings	radix sorts, tries, KMP, regexps, data compression	
advanced	B-tree, suffix array, maxflow	

Why study algorithms?

Their impact is broad and far-reaching.

Internet. Web search, packet routing, distributed file sharing, ...

Biology. Human genome project, protein folding, ...

Computers. Circuit layout, file system, compilers, ...

Computer graphics. Movies, video games, virtual reality, ...

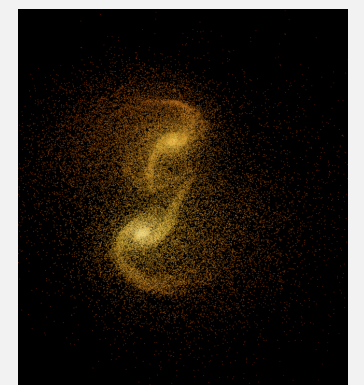
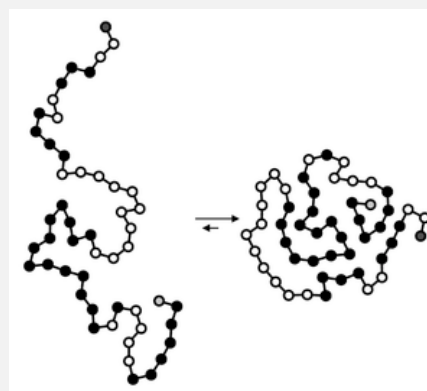
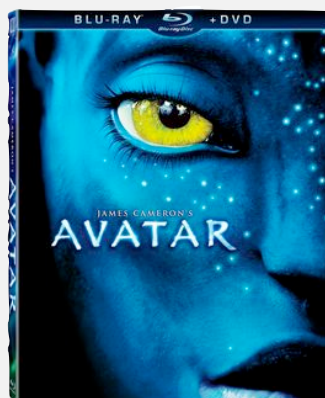
Security. Cell phones, e-commerce, voting machines, ...

Multimedia. MP3, JPG, DivX, HDTV, face recognition, ...

Social networks. Recommendations, news feeds, advertisements, ...

Physics. N-body simulation, particle collision simulation, ...

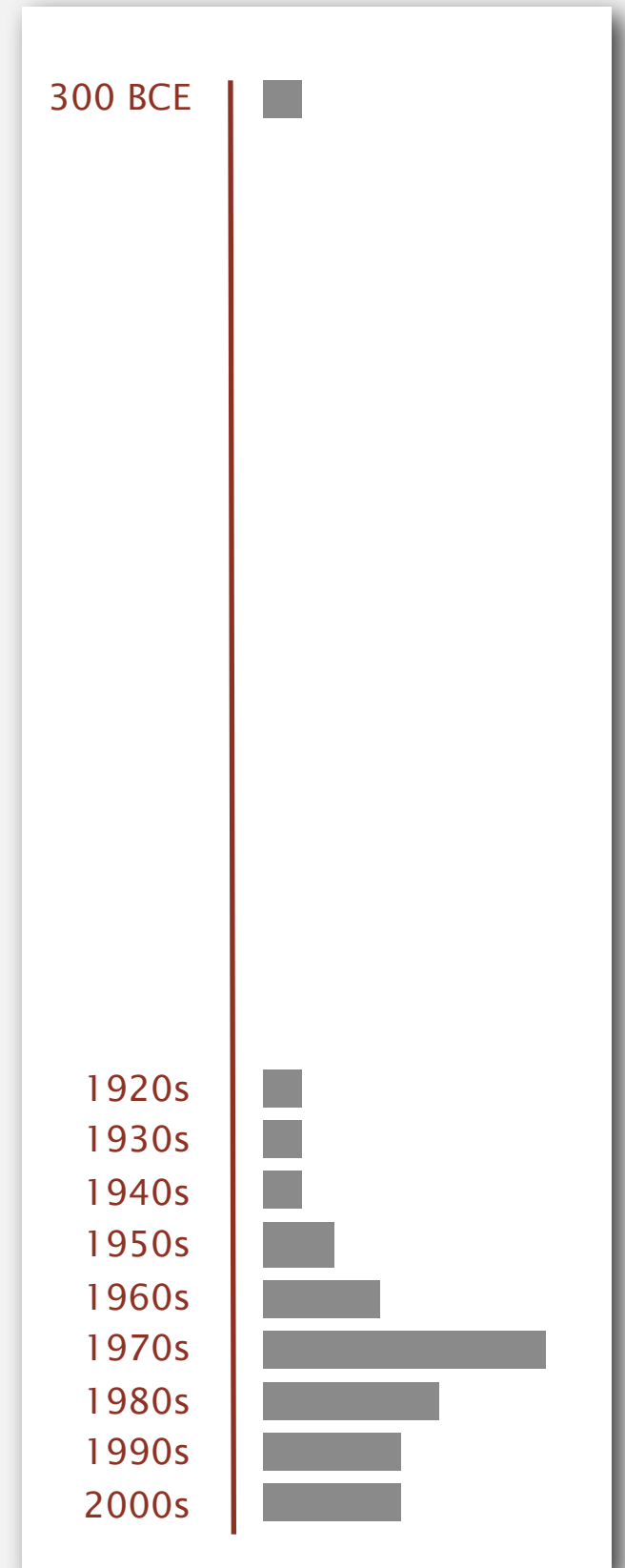
⋮



Why study algorithms?

Old roots, new opportunities.

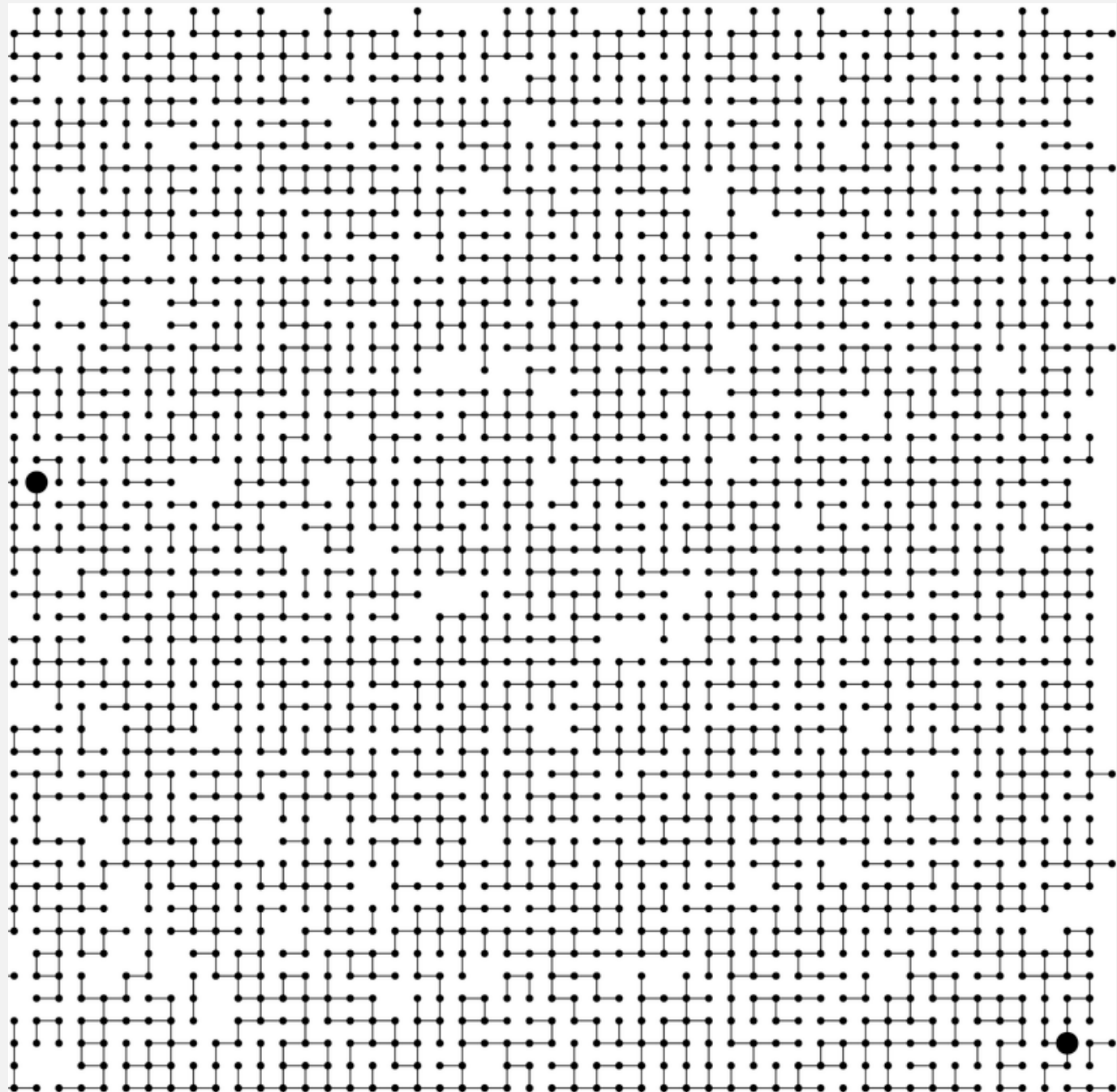
- Study of algorithms dates at least to Euclid.
- Formalized by Church and Turing in 1930s.
- Some important algorithms were discovered by undergraduates in a course like this!



Why study algorithms?

To solve problems that could not otherwise be addressed.

Ex. Network connectivity. [stay tuned]

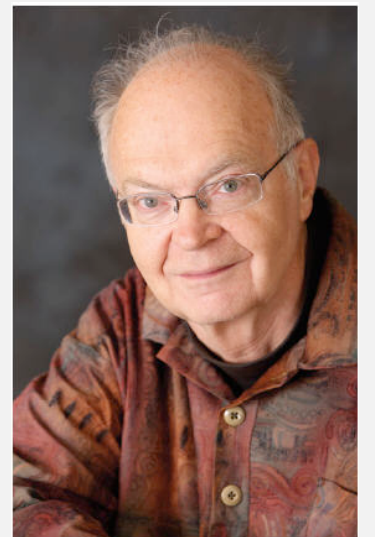
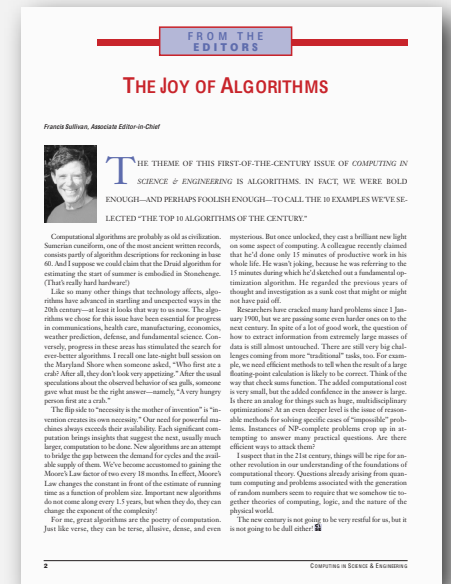


Why study algorithms?

For intellectual stimulation.

“ For me, great algorithms are the poetry of computation. Just like verse, they can be terse, allusive, dense, and even mysterious. But once unlocked, they cast a brilliant new light on some aspect of computing. ” — Francis Sullivan

“ An algorithm must be seen to be believed. ” — Donald Knuth



Why study algorithms?

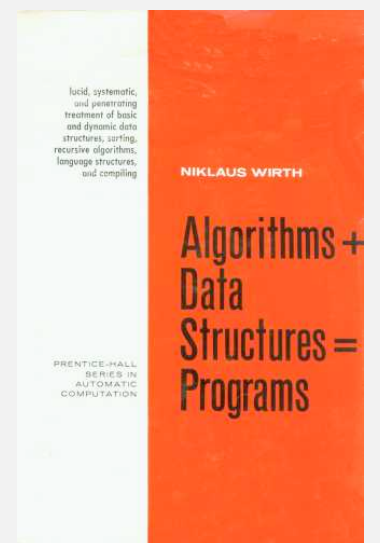
To become a proficient programmer.

“ I will, in fact, claim that the difference between a bad programmer and a good one is whether he considers his code or his data structures more important. Bad programmers worry about the code. Good programmers worry about data structures and their relationships. ”

— Linus Torvalds (creator of Linux)



“ Algorithms + Data Structures = Programs. ” — Niklaus Wirth



Why study algorithms?

They may unlock the secrets of life and of the universe.

Computational models are replacing math models in scientific inquiry.

$$\begin{aligned} E &= mc^2 \\ F &= ma \qquad F = \frac{Gm_1m_2}{r^2} \\ \left[-\frac{\hbar^2}{2m} \nabla^2 + V(r) \right] \Psi(r) &= E \Psi(r) \end{aligned}$$

20th century science
(formula based)

```
for (double t = 0.0; true; t = t + dt)
  for (int i = 0; i < N; i++)
  {
    bodies[i].resetForce();
    for (int j = 0; j < N; j++)
      if (i != j)
        bodies[i].addForce(bodies[j]);
  }
```

21st century science
(algorithm based)

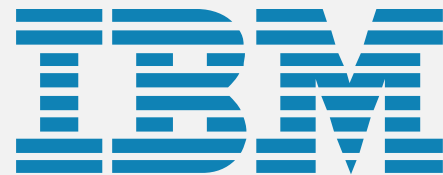
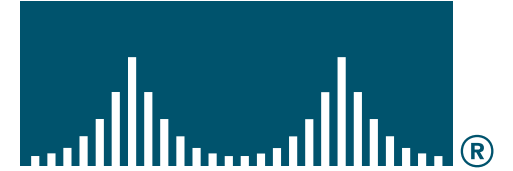
“ Algorithms: a common language for nature, human, and computer. ” — Avi Wigderson

Why study algorithms?

For fun and profit.



Apple Computer



Why study algorithms?

- Their impact is broad and far-reaching.
- Old roots, new opportunities.
- To solve problems that could not otherwise be addressed.
- For intellectual stimulation.
- To become a proficient programmer.
- They may unlock the secrets of life and of the universe.
- For fun and profit.

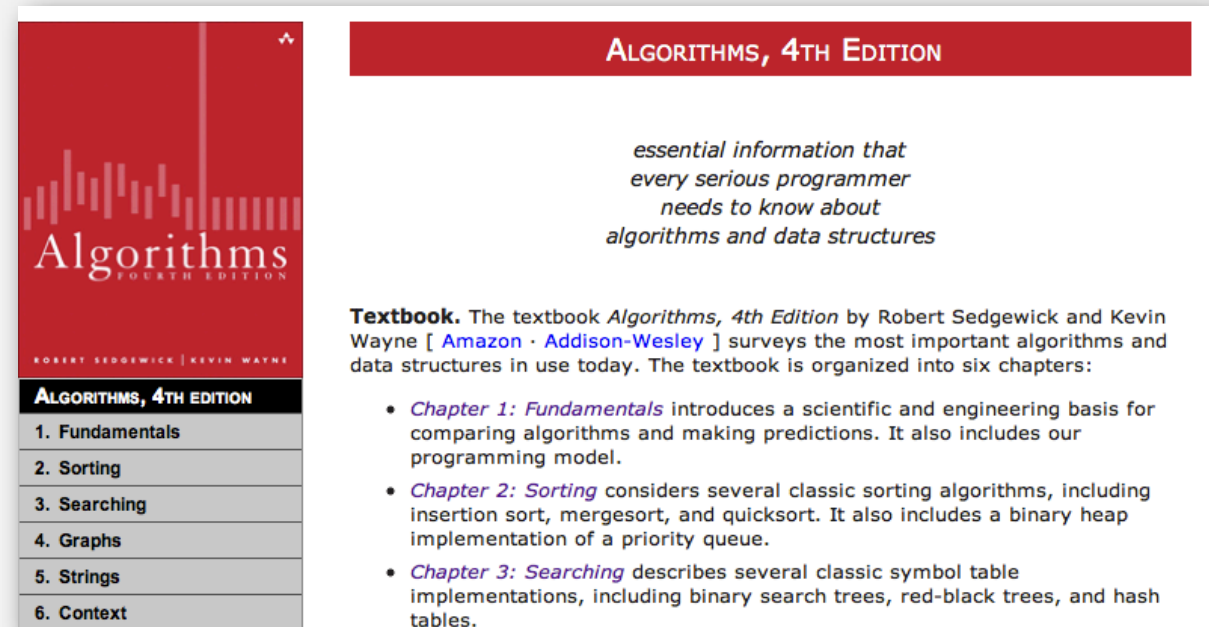
Why study anything else?



Resources

Booksite.

- Lecture slides.
- Download code.
- Summary of content.



The screenshot shows the website for 'Algorithms, 4th Edition'. On the left is a red book cover with the title 'Algorithms' and 'FOURTH EDITION' in white. Below the title is a list of six chapters: 1. Fundamentals, 2. Sorting, 3. Searching, 4. Graphs, 5. Strings, and 6. Context. On the right, the text 'ALGORITHMS, 4TH EDITION' is at the top. Below it is a quote: 'essential information that every serious programmer needs to know about algorithms and data structures'. Further down, a 'Textbook' section describes the book by Robert Sedgwick and Kevin Wayne, mentioning its availability on Amazon and Addison-Wesley, and listing the six chapters. Three bullet points provide details about the first three chapters: Chapter 1 (Fundamentals) covers scientific and engineering basis; Chapter 2 (Sorting) covers classic sorting algorithms like insertion sort, mergesort, and quicksort; Chapter 3 (Searching) covers classic symbol table implementations like binary search trees, red-black trees, and hash tables.

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Textbook (optional).

- *Algorithms, 4th edition* by Sedgwick and Wayne.
- More extensive coverage of topics.
- More topics.



ISBN 0-321-57351-X

Prerequisites

Prerequisites.

- Programming: loops, arrays, functions, objects, recursion.
- Java: we use as expository language.
- Mathematics: high-school algebra.

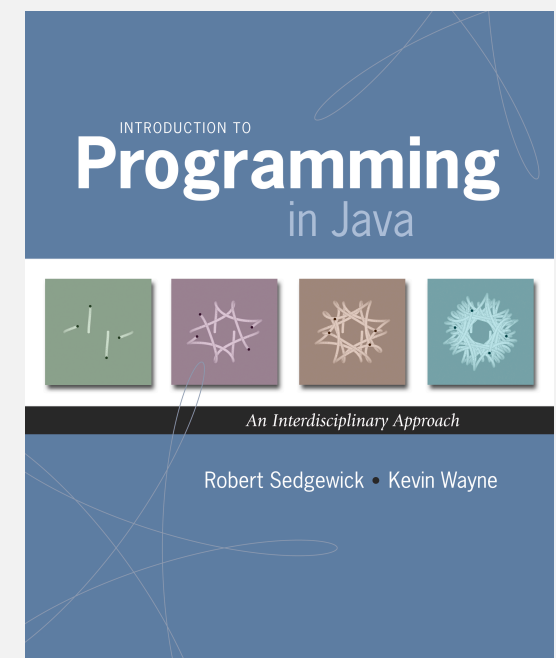
Review of prerequisite material.

- Quick: Sections 1.1 and 1.2 of *Algorithms, 4th edition*.
- In-depth: *An Introduction to programming in Java: an interdisciplinary approach* by Sedgewick and Wayne.

Programming environment.

- Use your own, e.g., Eclipse.
- Download ours (see instructions on web).

Quick exercise. Write a Java program.



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